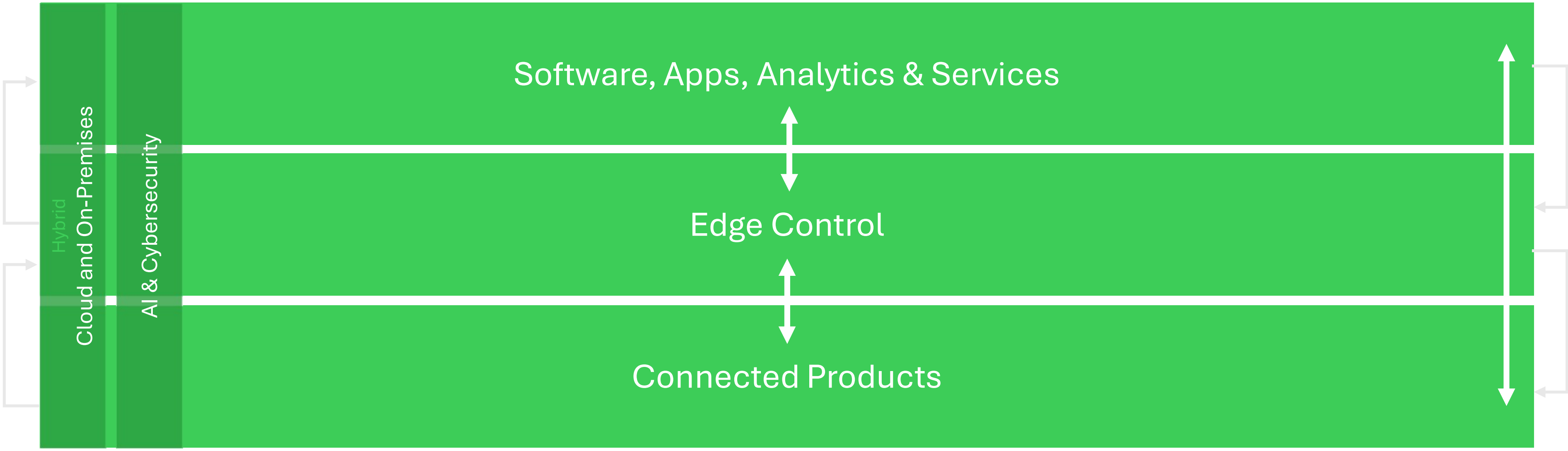
Power Products are parts of the EcoStruxure solutions

Energy Management, Operation efficiency



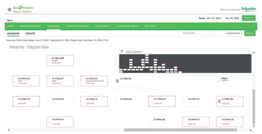
EcoStruxure™
Innovation At Every Level

<popup Software, Apps, Analytics & Services>

Analytics, Apps and Services

The data are collected and processed in the cloud-based EcoStruxure platform to provide advanced Analytics, Apps and Services.
The purpose is to enable advanced diagnostics, alarming, predictive and preventive maintenance...

EcoStruxure Energy Hub



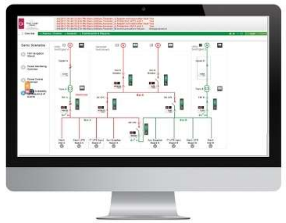
EcoStruxure Power Advisor

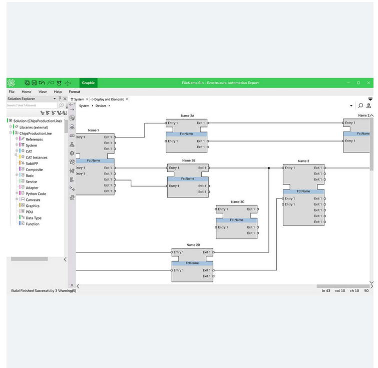
Connected Services Hub

<popup Edge Control>

Edge Control

Edge Control refers to the layer that provides local monitoring, control, and data processing at the edge of the network, close to the assets and processes.
Edge Control provide real-time data acquisition, visualisation, alarming, and control, ensuring fast, consistent access to actionable information for operators. Edge Control is essential for local decision-making, fault tolerance, and secure connectivity, while enabling integration with high level analytics and cloud services.
This layer consists of PLCs, controllers, SCADA systems, BMS, Automation systems, etc.



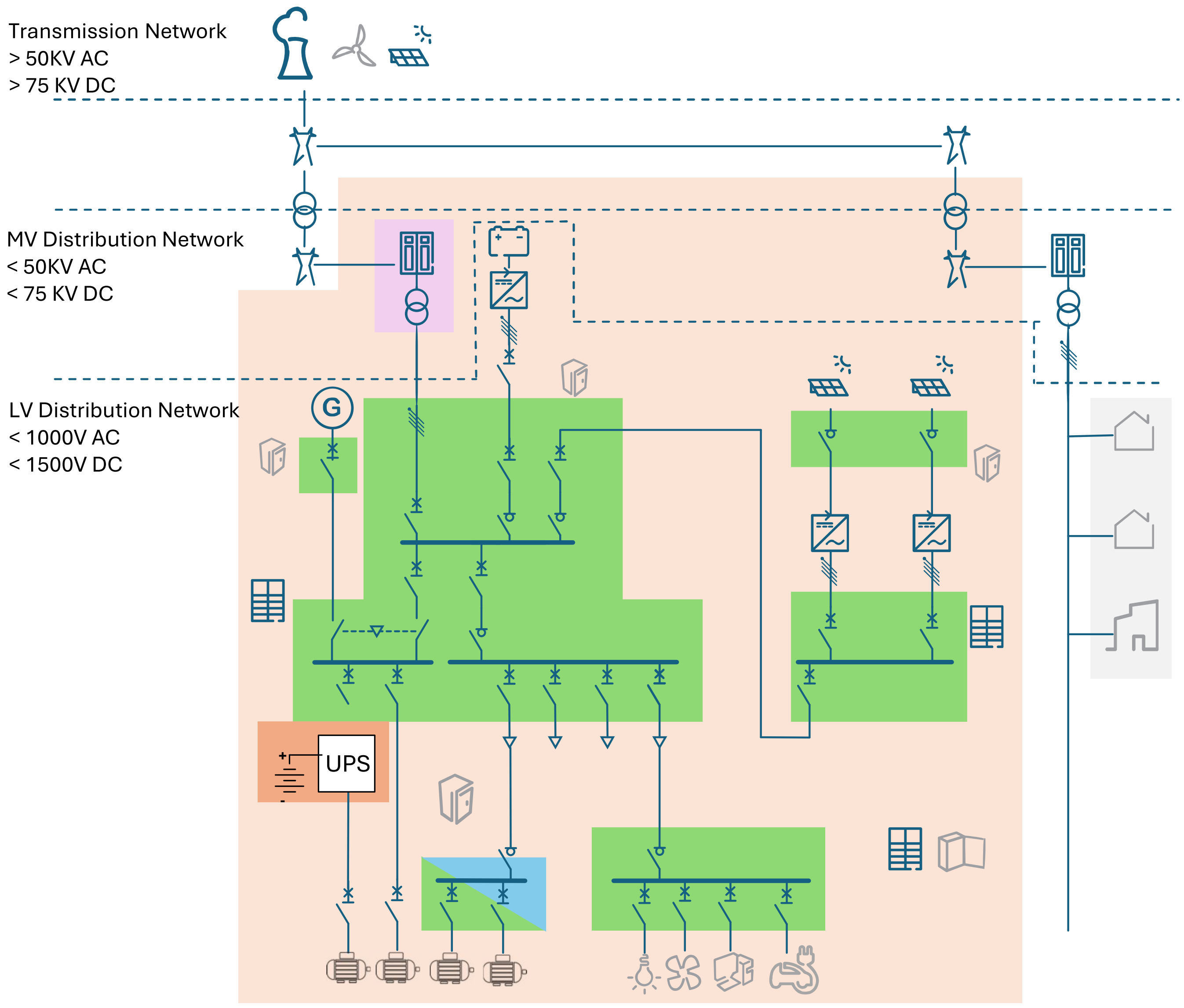


<popup Connected products>

Connected products

Connected products are capable to provide data regarding their statuses, health, electrical parameters through communication protocols (Modbus, Zigbee…)
They consist of advanced switchgears and controlgears (such as MasterPacT or ComPacT circuit breakers…), meters (ammeters, energy meters…), sensors (HeatTag…), gateways, UPS, EV Chargers…

Power Products in Electrical Architecture



The solution is mainly (but not exclusively) based on:

Power Products

Industrial Automation

Digital Energy

Power Systems

Secure Power

Home Solutions

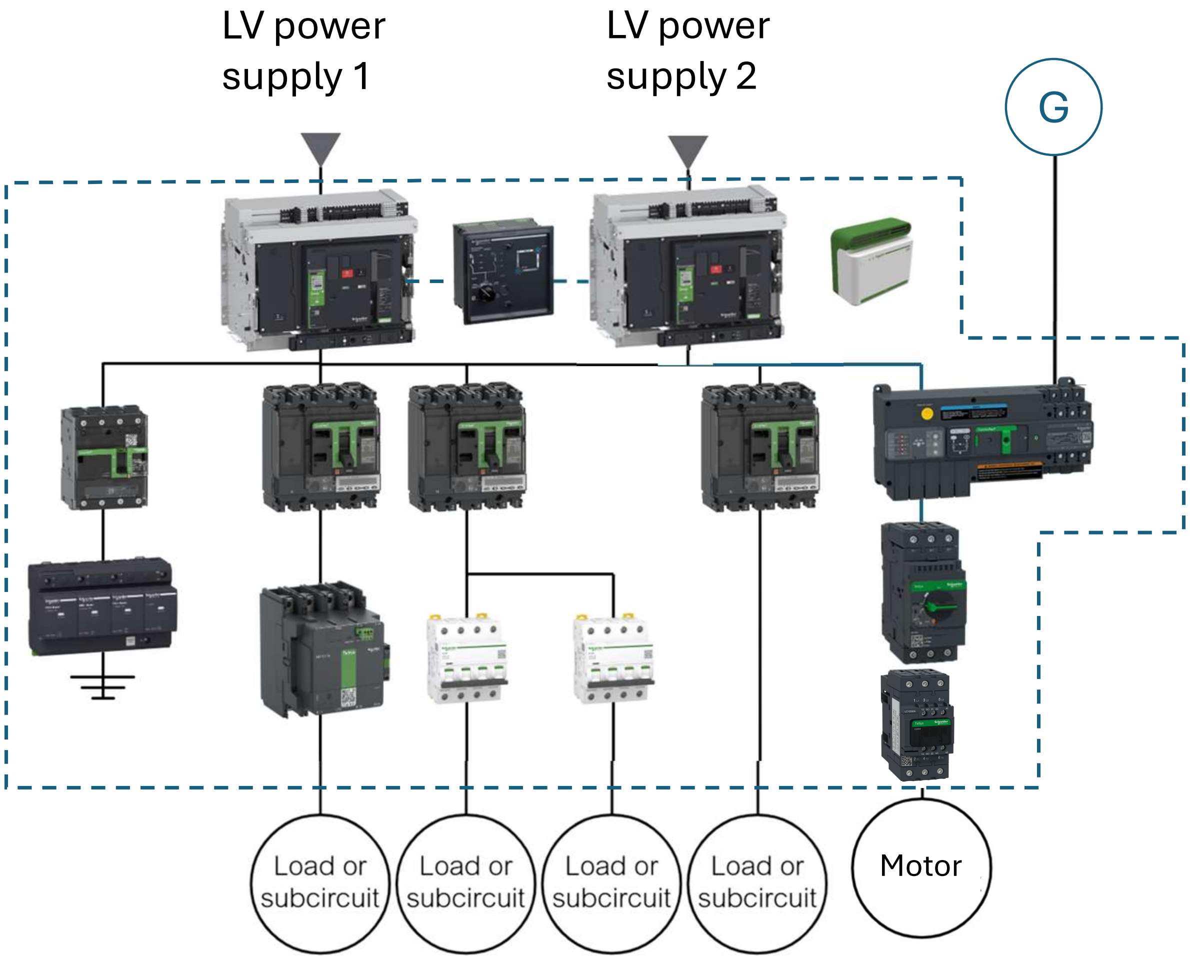
Power Products: What for?

Source Management

Isolation

Protection against:
- Short-circuit
- overload
- Earth-fault
- Surge…

Remote control

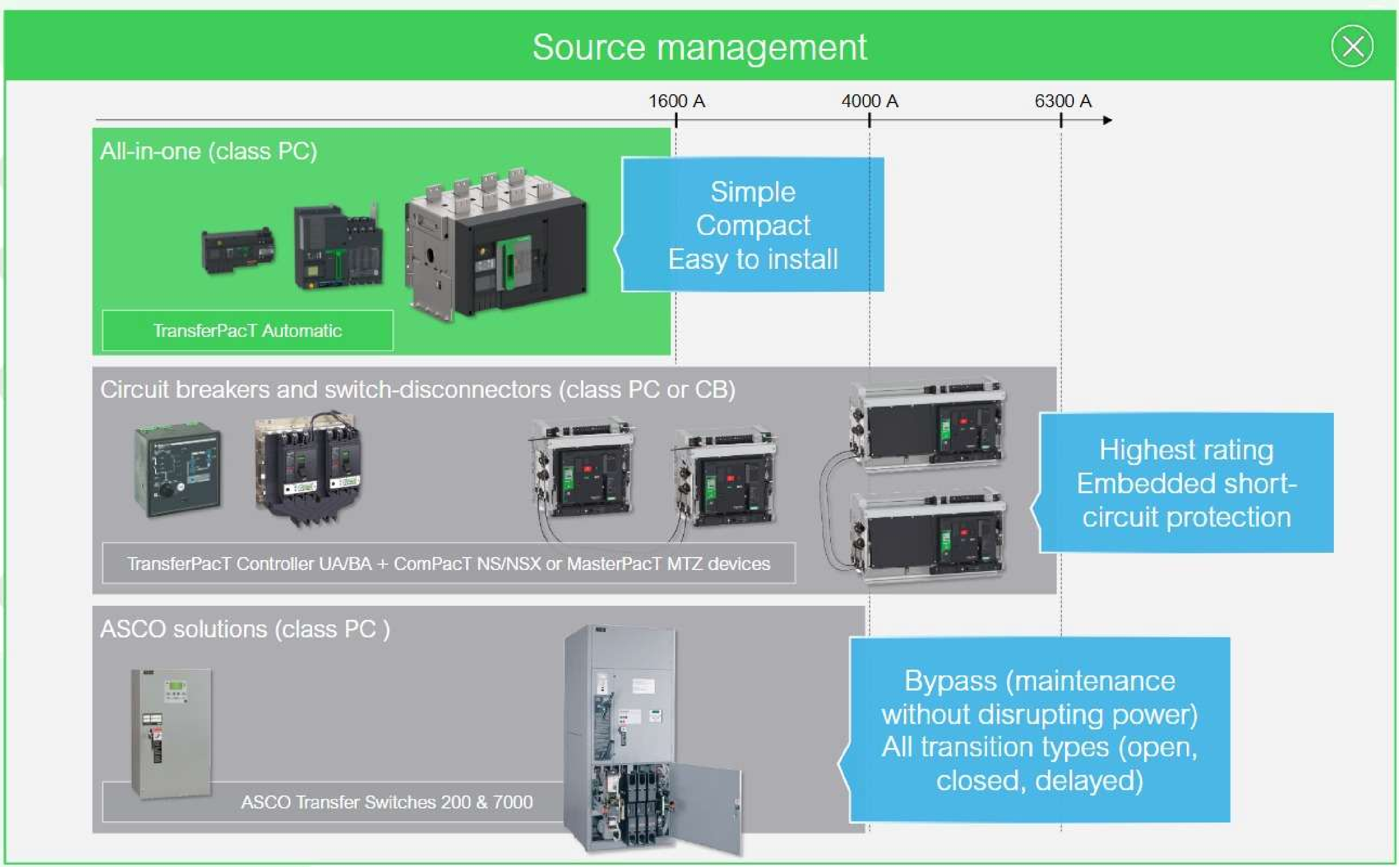


Monitoring and communication

Mechanical assembly and protection against live parts

Control gears and switchgears for safety, control and monitoring functions in LV architecture

<popup Source management>



<popup Isolation>

Isolation

In order to ensure operator safety during maintenance operations on the downstream circuit:

- The device shall ensure proper disconnection from voltage and current when in the off position.
- The device shall provide a clear indication of the power supply status.

Switch-disconnectors and circuit-breakers ensure isolation whatever the series:

- MasterPacT
- ComPacT
- TeSys Power, Vario, GS
- EasyPact
- Easy TeSys Power
- Acti9, Easy9
- Multi9
- Asco
- TransferPacT

<popup> Mechanical Assembly and protection against live parts.

Mechanical assembly and protection against live parts

X

Centralized distribution architecture

The devices are mechanically installed in switchboards, panels and final distribution enclosures (SeT series, Kaedra...).

The assembly should ensure protection:

- against live parts during operation and maintenance
- against arc flash risk...

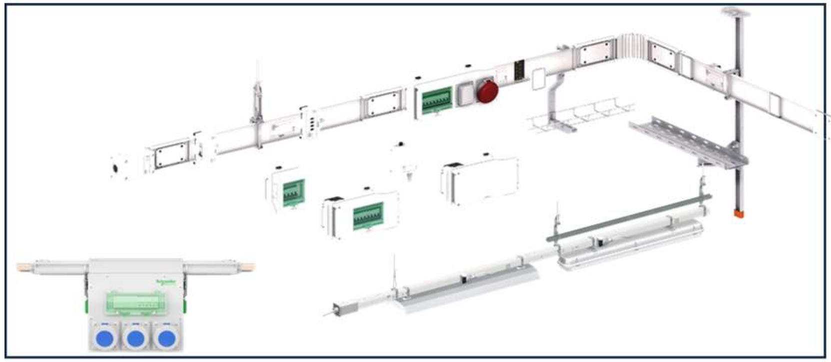
Schneider Electric switchboards and enclosures are also designed for continuity of service, easy installation, space saving...

Decentralized distribution architecture

The devices are mechanically installed in enclosures installed on busbar trunking system (Canalis, I-Line track). The lighting can also be directly installed on the busbar trunking system (Canalis KBA, KBB).

The busbar trunking system and tap off units ensure protection against live parts during operation and maintenance.

Busways also ensure high flexibility of the electrical distribution architecture. The enclosure (hot pluggable tap-off units) can be moved at any moment. The busways can also be dismounted and reused (unlike cables).



<popup> Mechanical Assembly and protection against live parts.

Remote control

X

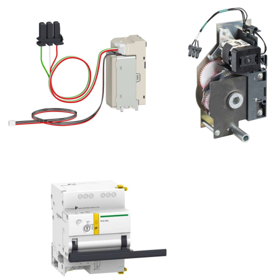
Different types of actuators are available to control remotely electrical distribution (to build the complete function the actuator can be associated with an HMI or automation system)

Contactors (TeSys, Easy TeSys, Acti9, Multi9)

The contactor is the device to remotely open and close an electrical circuit.
It does not ensure any electrical protection function (unlike circuit breaker or switch disconnector).
They are very common to control circuit of motors, heaters, lighting…

Electrical Auxiliaries for circuit breakers and switch disconnectors (Acti9, Multi9, PacT, EasyPact, TeSys Power…)

Electrical auxiliaries such as voltage releases (MN, MX and XF coils) and electrical motor (MCH) can be installed inside ACBs and MCCBs. For miniature circuit breakers the control system is installed on the side of the circuit breaker (Acti9 RCA).



Impulse relays (Acti9 iTL)

When at least 2 points of control are required.
Mainly for lighting.
No protection function unsured.

Other relays (Acti9 MIN…)

Specific relays enable remote control with additional automation function such as timers or automatic recloser

<popup> Mechanical Assembly and protection against live parts.

Monitoring and communication

X

Different types of actuators are available to control remotely electrical distribution (to build the complete function the actuator can be associated with an HMI or automation system)

Electronic control units (MicroLogic for PacT series)

The electronic control unit of PacT series circuit breakers enable monitoring of electrical parameters (current, voltage, energy…)

MicroLogic 7.0X for MasterPacT MTZ

MicroLogic 7.2 for ComPacT NSX

Gateways (Enerlin'X, Acti9)

The gateways enable connections of devices equipped with MicroLogic control units or devices equipped with indication contacts to Modbus TCP or Modbus SL networks

Indication Auxiliaries (Acti9, Multi9, PacT, EasyPact, TeSys…)

Indication auxiliaries (OF, SD…) installed inside or on the side of devices enable reporting of the status (open, closed, tripped…) They can be connected to indication lights or automation systems.

Indication contacts (wireless and wired for ComPacT NSX)

Acti9 iOF indication contact (installed on DIN rail on the side of Acti9 CB or switch disconnector)

All-in-one devices (Acti9 Reflex, Active)

Some devices Acti9 Reflex iC60 or Acti9 Active embeds indication functions (wired or wireless).

<popup> Protection

Protection

X

	Protection functions (not exhaustive)	 Circuit breaker	 Fuse holder	 Switch disconnector	 Overload protection relay	 Surge Protection Devices (SPD)
Basic functions	Short-circuit protection	X	X			
	Overload protection	(X)	X		X	
	Isolation	X	(X)	X		
	Earth-fault protection	(X)		(X)		
Additional functions	Under and overvoltage protection	(X)			(X)	
	Final distribution Arc fault protection	(X)				
	Transient overvoltage protection					X

X : The function is ensured whatever the reference of the Schneider Electric device

(X) The function can be ensured depending on the reference and/or optional auxiliary/addon

<popup> Protection circuit breaker>

Protection

X

	Protection functions (not exhaustive)	 Circuit breaker	 Fuse holder	 Switch disconnector	 Overload protection relay	 Surge Protection Devices (SPD)
Basic functions	Short-circuit protection	x	x			
	Overload protection		x		x	
	Isolation		(X)	x		
	Earth-fault protection			(X)		
Additional functions	Under and overvoltage protection					
	Final distribution Arc fa protection					
	Transient overvoltage p					x

Short-circuit protection

What:
Open the circuit as fast as possible to protect cable and avoid fire in case of short-circuit.

How:
This function can be ensured either by circuit breakers or fuses.
However, compared to fuses circuit breakers allow for:
- minimized downtime as they can be rearmed after the fault has been cleared. (The fuse must be replaced).
- more flexibility: for instance MicroLogic control units of PacT series allows various settings regarding protection (no settings for the fuses).

X : The function is ensured whatever the reference of the Schneider Electric device

(X) The function can be ensured depending on the reference and/or optional auxiliary/addon

<popup> Protection >

Protection

X

	Protection functions (not exhaustive)	 Circuit breaker	 Fuse holder	 Switch disconnector	 Overload protection relay	 Surge Protection Devices (SPD)
Basic functions	Short-circuit protection +	x	x			
	Overload protection +	(X)	x		x	
	Isolation	x	(X)	x		
	Earth-fault protection			(X)		
Additional functions	Under and overvoltage protection					
	Final distribution Arc fa protection					
	Transient overvoltage p					x

Overload protection

What:
Open the circuit to protect loads and cables (but not to early to avoid nuisance tripping)
.
How:
In electrical distribution architecture this function can be ensured either by circuit breakers or fuses.
In motor starters, this function can be ensured either by circuit breakers, or fuses or overload thermal relays..

X : The function is ensured whatever the reference of the Schneider Electric device

(X) The function can be ensured depending on the reference and/or optional auxiliary/addon

<popup> Protection

Protection

X

	Protection functions (not exhaustive)	 Circuit breaker	 Fuse holder	 Switch disconnector	 Overload protection relay	 Surge Protection Devices (SPD)
Basic functions	Short-circuit protection	X	X			
	Overload protection	(X)	X		X	
	Isolation	X	(X)	X		
	Earth-fault protectio[n]			(X)		
Additional functions	Under and overvoltag[e] protection					
	Final distribution Arc protection					
	Transient overvoltage					X

X : The function is [...] Schneider Electric device

(X) The function can be ensured depending on the reference and/or optional auxiliary/addon

Isolation

What:
Make sure that the circuit is not live during maintenance
.
How:
All the circuit breakers from Schneider Electric ensure isolation function.
Some fuse holders can also ensure isolation function.

<popup> Protection

Protection

X

Earth fault protection

What:
Open the circuit to protect people against electrocution and/or avoid fire and damages on loads
.
How:
There are different types of circuit breakers which can ensure this function. It can depend on:
- the reference of the breaker (Acti9, Easy9)
- the control unit (PacT series)
- add-ons (Vigi Add-On for Acti9, ComPacT NSX)

	Protection functions		 Fuse holder	 Switch disconnector	 Overload protection relay	 Surge Protection Devices (SPD)
Basic functions	Short-circuit protectio[n]		X			
	Overload protection		X		X	
	Isolation		(X)	X		
	Earth-fault protection +	(X)		(X)		
Additional functions	Under and overvoltage protection +	(X)				
	Final distribution Arc fault protection +	(X)				
	Transient overvoltage protection +					X

X : The function is ensured whatever the reference of the Schneider Electric device

(X) The function can be ensured depending on the reference and/or optional auxiliary/addon

<popup> Protection

Protection

X

	Protection functions		 Fuse holder	 Switch disconnector	 Overload protection relay	 Surge Protection Devices (SPD)
Basic functions	Short-circuit protectio		x			
	Overload protection		x		x	
	Isolation		(X)	x		
	Earth-fault protection			(X)		
Additional functions	Under and overvoltage protection	(X)				
	Final distribution Arc fault protection	(X)				
	Transient overvoltage protection					x

Under and overvoltage protection

What:
Open the circuit to protect sensitive loads and electrical installation (but not too early to avoid nuisance tripping)
.
How:
Circuit breakers of the PacT series can ensure this function thanks to MicroLogic control unit.
Acti9 Active RCBOs also features the overvoltage protection.
Some.
In motor starters, this function can also be ensured by electronic thermal overload relays (TeSys Tera, TeSys Giga, TeSys Deca advanced)

X : The function is ensured whatever the reference of the Schneider Electric device

(X) The function can be ensured depending on the reference and/or optional auxiliary/addon

<popup> Protection

Protection

X

	Protection functions		 Fuse holder	 Switch disconnector	 Overload protection relay	 Surge Protection Devices (SPD)
Basic functions	Short-circuit protectio		x			
	Overload protection		x		x	
	Isolation		(X)	x		
	Earth-fault protection			(X)		
Additional functions	Under and overvoltage protection					
	Final distribution Arc fault protection	(X)				
	Transient overvoltage protection					x

Final distribution arc fault protection

What:
Open the circuit as fast as possible to avoid fire in case of Arc Fault (at the socket level for instance).
.
How:
Acti9 Active RCBOs or Acti9 Acvie VigiARC add-ons.

X : The function is ensured whatever the reference of the Schneider Electric device

(X) The function can be ensured depending on the reference and/or optional auxiliary/addon

<popup> Protection

Protection

X

Protection functions (not exhaustive)		 Circuit breaker	 Fuse holder	 Switch disconnector	 Overload protection relay	 Surge Protection Devices (SPD)
Basic functions	Short-circuit protectio		x			
	Overload protection		x		x	
	Isolation		(X)	x		
	Earth-fault protection			(X)		
Additional functions	Under and overvoltage protection				(X)	
	Final distribution Arc fa protection					
	Transient overvoltage protection +					x

Transient overvoltage protection

What:
Evacuate the surge (due to lightning for instance) to Earth in order to protect sensitive loads and electrical installation
.
How:
Acti9 Surge Protection Devices (SPD).

X : The function is ensured whatever the reference of the Schneider Electric device

(X) The function can be ensured depending on the reference and/or optional auxiliary/addon

<popup> Mechanical Assembly and protection against live parts.

Protection

X

Event	Goal of the protection	Circuit breaker	Fuse holder	Switch disconnector	Overload protection relay	Surge Protection Devices (SPD)
Short-circuit	Open the circuit as fast as possible to protect cable and avoid fire	X	X			
Overload	Open the circuit to Protect loads and cables (but not to early to avoid nuisance tripping)	(X)	X		X	
Isolation	Make sure that the circuit is not live during maintenance	X	(X)	X		
Earth-fault	Protect people and/or avoid fire and damages on loads	(X)		(X)		
Under and overvoltage	Open the circuit to protect sensitive loads and electrical installation but not too early to avoid nuisance tripping	(X)				
Final distribution Arc fault	Open the circuit as fast as possible to avoid fire	(X)				
Transient overvoltage (Surge due to Lightning…)	Evacuate the surge to earth in order to protect sensitive loads and electrical installation					X

(X) The function can be ensured depending on the reference and/or optional auxiliary/addon

X : The function is ensured whatever the reference of the Schneider Electric device